

Ema Lovrić, 16.4.2026.

Assignment 2: Local office communication system

<https://github.com/elovric13/eutech.git>

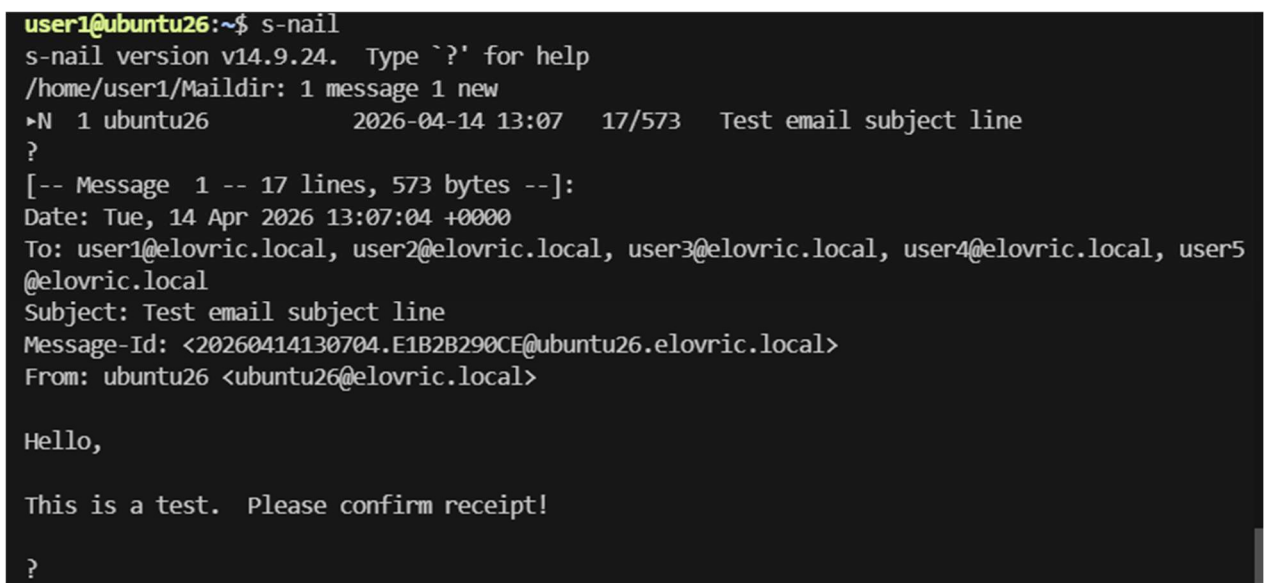
Postfix configuration

On Ubuntu VM, the local mail server is configured using Postfix. The configuration involves setting up local user accounts, aliases, and forwarders to manage mail server. Test emails were sent locally between the created user accounts using the command line mail client (s-nail). Additional tests confirmed that emails sent to the alias were delivered to right user, and emails sent to the forwarder were successfully routed to the external Outlook address.

1. Install and configure local mail server: Postfix was installed and configured as an "Internet Site". The system mail name was set to elovric.local. The main configuration file (/etc/postfix/main.cf) was updated to define the domain and network settings.

```
home_mailbox = Maildir/  
virtual_alias_maps = hash:/etc/postfix/virtual  
virtual_alias_domains = elovric.local
```

2. Create user email accounts: Five local Linux user accounts (user1, user2, user3, user4, user5) were created using the adduser command. These accounts serve as the local mailboxes for receiving test emails.



```
user1@ubuntu26:~$ s-nail  
s-nail version v14.9.24. Type `?' for help  
/home/user1/Maildir: 1 message 1 new  
➤N 1 ubuntu26      2026-04-14 13:07   17/573   Test email subject line  
?  
[-- Message 1 -- 17 lines, 573 bytes --]:  
Date: Tue, 14 Apr 2026 13:07:04 +0000  
To: user1@elovric.local, user2@elovric.local, user3@elovric.local, user4@elovric.local, user5@elovric.local  
Subject: Test email subject line  
Message-Id: <20260414130704.E1B2B290CE@ubuntu26.elovric.local>  
From: ubuntu26 <ubuntu26@elovric.local>  
  
Hello,  
  
This is a test. Please confirm receipt!  
  
?
```

Figure 1: Test email sent locally between multiple users

3. Configure aliases, forwarders and attachment size limit. Aliases and forwarders were configured in the /etc/postfix/virtual file.
 - Aliases:
 - info@elovric.local was mapped to user1
 - support@elovric.local was mapped to user2

```

user2@ubuntu26:~$ s-nail
s-nail version v14.9.24.  Type `?' for help
/home/user2/Maildir: 3 messages 1 new 2 unread
  O 1 ubuntu26          2026-04-14 13:07   17/573   Test email subject line
  U 2 info@elovric.local 2026-04-14 16:14   16/451   Test aliases
>N 3 info@elovric.local 2026-04-14 16:15   16/466   Test aliases!
?
[-- Message 3 -- 16 lines, 466 bytes --]:
Date: Tue, 14 Apr 2026 16:15:12 +0000
From: info@elovric.local
To: support@elovric.local
Subject: Test aliases!
Message-ID: <20260414161512.0J20d%info@elovric.local>

Hello,

This is message for testing aliases.

? █

```

Figure 2: Test email sent to the alias info@elovric.local

- Forwarders: forwarding emails from a local address to an external email address.
 - user5@elovric.local forwarded to elovric@algebra.hr

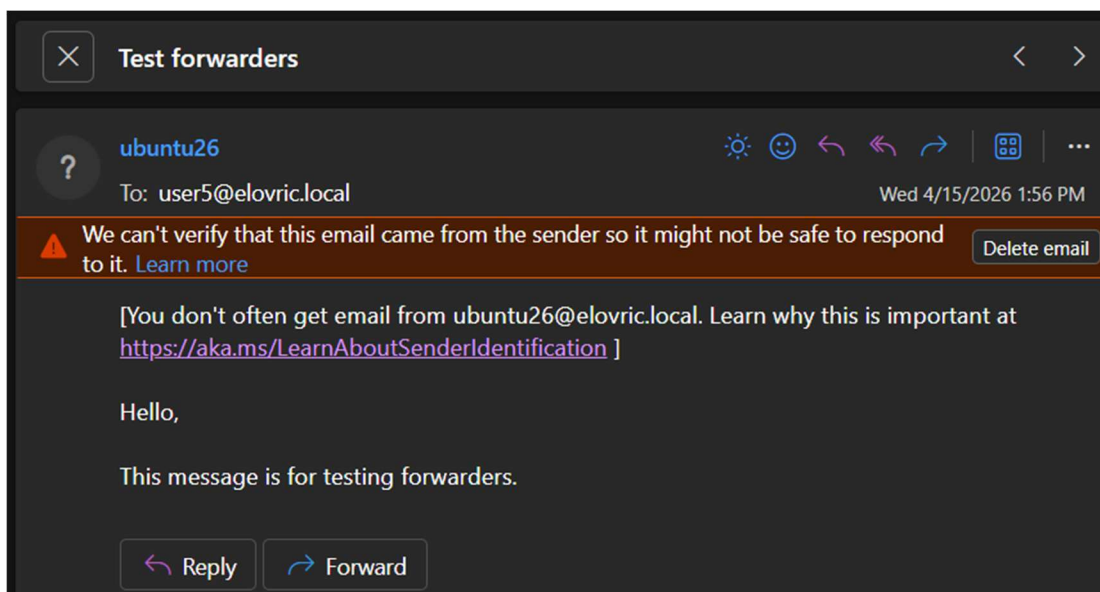


Figure 3: Test email sent to the forwarder (user5@elovric.local) and received in Outlook

- Attachment size limit: The maximum allowed email attachment size was configured by modifying the `message_size_limit` parameter in the Postfix `main.cf` file

```
home_mailbox = Maildir/  
virtual_alias_maps = hash:/etc/postfix/virtual  
virtual_alias_domains = mydomain.local  
message_size_limit = 5242880  
"/etc/postfix/main.cf" [readonly] 51L, 1520B
```

Figure 4: Configuration of message_size_limit in main.cf

File sharing and printers

1. Set up file sharing: Samba was installed and configured to act as a file server. Shared directories (/srv/smbashare/[admin, it, hr, public]) were created, and the /etc/samba/smb.conf file was updated to define shared directories, making it browsable on the local network.

```
[admin]  
comment = Admin folder for user ubuntu26  
path = /srv/smbashare/admin  
browsable = yes  
guest ok = no  
read only = no  
create mask = 0755  
valid users = @admin  
  
[it]  
comment = IT folder for users user1, user2  
path = /srv/smbashare/IT  
browsable = yes  
guest ok = no  
read only = no  
create mask = 0755  
valid users = @it  
write list = @it  
  
[hr]  
comment = HR folder for users user2, user3  
path = /srv/smbashare/HR  
browsable = yes  
guest ok = no  
read only = no  
create mask = 0755  
valid users = @hr  
  
[public]  
comment = Shared folder for everyone  
path = /srv/smbashare/public  
browsable = yes  
guest ok = yes  
read only = no  
create mask = 0755
```

Figure 1: Samba share configuration

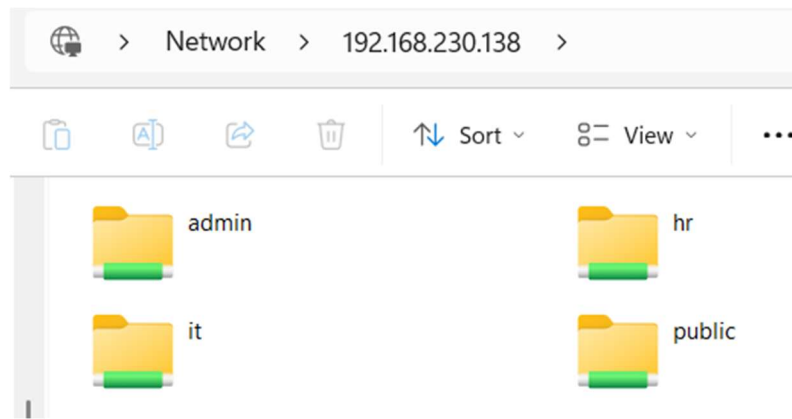


Figure 2: Samba share visible from a client machine on the network

2. Role-based access: Role-based access was implemented using Samba groups. Users were placed in specific groups and given permissions to access each directory. Tests confirmed that some users can access folders and have right permissions, while others received an 'Access denied' error.

```
ubuntu26@ubuntu26:~$ smbclient //192.168.230.138/public
Password for [WORKGROUP\ubuntu26]:
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \>
```

Figure 3: Anonymous access to public share

```
ubuntu26@ubuntu26:~$ smbclient //192.168.230.138/admin -U user1
Password for [WORKGROUP\user1]:
tree connect failed: NT_STATUS_ACCESS_DENIED
ubuntu26@ubuntu26:~$ smbclient //192.168.230.138/hr -U user1
Password for [WORKGROUP\user1]:
tree connect failed: NT_STATUS_ACCESS_DENIED
```

Figure 4: Unauthorized user receiving an 'Access denied' error when attempting to connect

```

ubuntu26@ubuntu26:~$ smbclient //192.168.230.138/hr -U user2
Password for [WORKGROUP\user2]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D            0   Thu Apr 16 11:12:17 2026
..               D            0   Thu Apr 16 11:12:17 2026
hr.txt           N            0   Thu Apr 16 10:59:31 2026

ubuntu26@ubuntu26:~$ smbclient //192.168.230.138/hr -U user2
Password for [WORKGROUP\user2]:
Try "help" to get a list of possible commands.
smb: \> put user2_test.txt
putting file user2_test.txt as \user2_test.txt (0.0 kb/s) (average 0.0 kb/s)
smb: \> ls
.                D            0   Thu Apr 16 11:49:51 2026
..               D            0   Thu Apr 16 11:49:51 2026
user2_test.txt   A            0   Thu Apr 16 11:49:51 2026
hr.txt           N            0   Thu Apr 16 10:59:31 2026

10218772 blocks of size 1024. 2147512 blocks available
smb: \>

```

Figure 5: : Authorized user successfully accessing and writing to the restricted share

3. Share printers over the local network: The Common UNIX Printing System (CUPS) was installed to manage and share printers. A virtual PDF printer was installed and shared. The CUPS configuration file (/etc/cups/cupsd.conf) was modified to listen on the server's LAN IP address, making the printer available to other machines.

```

# Restrict access to the server...
<Location />
    Order allow,deny
    Allow localhost
    Allow 192.168.230.*
</Location>

# Restrict access to the admin pages...
<Location /admin>
    Order allow,deny
    Allow localhost
    Allow 192.168.230.*
</Location>

# Restrict access to configuration files...
<Location /admin/conf>
    Order allow,deny
    Allow localhost
    Allow 192.168.230.*
</Location>

# Restrict access to log files...
<Location /admin/log>
    AuthType Default
    Require user @SYSTEM
    Order allow,deny
</Location>

```

Figure 6: CUPS configuration

OpenPrinting CUPS Home Administration Classes Help Jobs Printers

PDF

PDF (Idle, Accepting Jobs, Shared, Server Default)

Maintenance Administration

Description: PDF

Location:

Driver: Generic CUPS-PDF Printer (w/ options) (color)

Connection: cups-pdf/

Defaults: job-sheets=none, none media=unknown sides=one-sided

Jobs

Search in PDF: Search Clear

Show Active Jobs Show All Jobs

Completed jobs in order of completion or cancellation ▲

ID	Name	User	Size	Pages	State	Control
PDF-6	Unknown	Withheld	1k	1	completed at Thu 16 Apr 2026 11:59:39 AM UTC	Reprint Job

Figure 7: CUPS web interface with the shared PDF printer

- Configure shared printer access: Printer access was controlled at the network level. The `<Location />` block in the CUPS configuration was updated to 'Allow 192.168.230.*' (the local subnet), ensuring that only machines on the local network could submit print jobs or view the printer status.

Bluetooth & devices > Printers & scanners > PDF @ ubuntu26



PDF @ ubuntu26

Printer status: Idle

[PDF @ ubuntu26 settings](#)

Figure 8: Added printer on Windows machine

← → ↻ Nije sigurno 192.168.230.138:631/printers/PDF?... ☆

ABOD robCov RNN OCSVM 0x20 Institutions español >> Sve oznake

OpenPrinting CUPS Home Administration Classes Help Jobs **Printers**

PDF

PDF (Idle, Accepting Jobs, Shared, Server Default)

Maintenance ▾ Administration ▾

Description: PDF

Location:
Driver: Generic CUPS-PDF Printer (w/ options) (color)

Connection: cups-pdf:/

Defaults: job-sheets=none, none media=unknown sides=one-sided

Jobs

Search in PDF: Search Clear

Show Active Jobs Show All Jobs

Completed jobs in order of completion or cancellation ▲

ID	Name	User	Size	Pages	State	Control
PDF-5	Test Page	anonymous	1k	1	completed at Thu Apr 16 11:52:13 2026	Reprint Job

Figure 9: Shared printer visible from a client machine's browser

Email manual: Command-line (s-nail)

You must be logged into the server using your assigned Linux user account (e.g., user1).

To send an email to another user on the local network, follow these steps:

1. Open your terminal and type the following command, replacing the email address with your recipient:
`s-nail user2@elovric.local`
2. The system will prompt you for a Subject. Type your subject line and press Enter.
3. Type the body of your message.
4. When you are finished typing your message, press Enter to go to a new blank line, then press CTRL+D to send the email.

```
ubuntu26@ubuntu26:~$ s-nail user2@elovric.local
Subject: Test message
To: user2@elovric.local
Dear user2,

This is a test message.

Kind regards
^D
-----
(Preliminary) Envelope contains:
To: user2@elovric.local
Subject: Test message
Send this message [yes/no, empty: recompose]? yes
```

To check your inbox and read received emails:

1. In a terminal, type the command: `s-nail`
2. You will see a list of your emails, each with a number next to it. The "N" indicates a New message
3. To read a specific email, type its number (e.g., 1)
4. To return to the message list, type "h"
5. To delete the email you just read, type "d"
6. To exit the mail client, type "q" (quit)

```
ubuntu26@ubuntu26:~$ s-nail
s-nail version v14.9.24. Type `?' for help
/home/ubuntu26/Maildir: 2 messages 1 unread
>U 1 info@elovric.local 2026-04-14 16:12 16/460 Test aliases
  0 2 info@elovric.local 2026-04-14 16:13 16/446 Test aliases
? 2
[-- Message 2 -- 16 lines, 446 bytes --]:
Date: Tue, 14 Apr 2026 16:13:29 +0000
From: info@elovric.local
To: test@elovric.local
Subject: Test aliases
Message-ID: <20260414161329.09tL_%info@elovric.local>

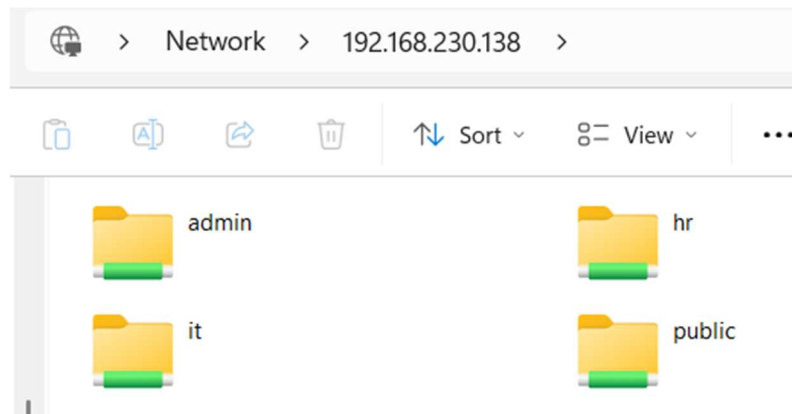
Hello,

This is message for testing aliases.
```


Shared resource manual

a. Accessing shared folders (Windows)

1. Open File Explorer
2. Enter Server Address:
<\\192.168.230.138>
3. Enter Credentials
4. Access folder and files. You will only be able to open folders you have permission to access.



b. Connecting to the shared printer (Windows)

1. Open Settings, navigate to Printers.
2. Click the "Add device" button.
3. Add Manually: Wait a few seconds, then click "Add manually" next to "The printer that I want isn't listed".
4. Select shared printer: Select shared printer by name. Enter printer URL:
<http://192.168.230.138:631/printers/PDF>
5. Install Driver to complete the installation.

Bluetooth & devices > Printers & scanners > PDF @ ubuntu26



PDF @ ubuntu26

Printer status: Idle

[PDF @ ubuntu26 settings](#)